

INTRODUCTION TO DATA SCIENCE

CO-1 : Exam Notes

B.Tech CSE — Semester VII | PTU Syllabus

Evolution, Need & Components of Data Science · Data Science Process · Data Science vs Business Intelligence · Applications · Ethical Issues, Privacy, Security & Responsible Data Use

Learning Objectives of CO-1

After studying this unit, a student should be able to:

- Explain how data science evolved as a discipline and why it is needed today
- Identify and describe the core components of data science
- Describe, stage by stage, the data science process
- Distinguish data science from business intelligence with examples
- List and explain applications of data science in different fields
- Discuss the ethical, privacy, security and responsible-use dimensions of data science practice

1. Evolution of Data Science

1.1 What is Data Science?

Data Science is an interdisciplinary field that combines scientific methods, statistical techniques, computational algorithms, and systems to extract knowledge, patterns, and actionable insight from data — whether structured (rows/columns, e.g. spreadsheets, relational DBs) or unstructured (text, images, audio, video, sensor streams, social media).

Definition to remember: “Data science is the discipline of extracting knowledge and insight from data, in various forms, using scientific methods, processes, algorithms and systems, in order to support better decision-making.”

It is interdisciplinary because no single subject (statistics, computer science, or business management) alone covers what a data scientist does — it draws on all of them together (see Section 3).

1.2 Historical Evolution — Stage by Stage

For 'trace the evolution' questions, write all 6 stages in order, stating (a) dominant technology and (b) main purpose of data at each stage.

Stage 1 — Data Processing Era (before 1990s)

- Computers used mainly for record-keeping & transaction processing (payroll, inventory, accounting) in flat files / early databases
- Purpose: purely operational — keep business running, not generate insight
- Analysis was manual (clerks/accountants), limited to simple totals

Stage 2 — Database & Data Warehousing Era (1990s)

- RDBMS matured (relational model + SQL) → larger, structured datasets, flexible querying
- **Data warehouse:** large central repository consolidating data from many operational systems (sales, finance, HR) for querying/reporting, not transactions
- Shift: data as operational record → data as a resource for management reporting

Stage 3 — Business Intelligence Era (late 1990s–2000s)

- Dashboards, scorecards, OLAP cubes allow slicing/dicing historical data (e.g. sales by region/product/month)
- Gave rise to the 'data-informed manager' checking KPI dashboards
- BI is fundamentally descriptive — tells what happened / is happening, not the future

Stage 4 — Big Data Era (2000s–2010s)

- Internet, e-commerce, mobile, social media, IoT → explosion of data beyond traditional DB/BI capability

The 3 V's:

- **Volume:** sheer quantity of data (terabytes/petabytes)
- **Velocity:** speed of data generation/processing, sometimes real-time (stock trades, sensors)
- **Variety:** many forms — tables, text, images, audio, video, logs

(Extended to 4th/5th V: Veracity = trustworthiness/quality; Value = usefulness of insight.) New distributed storage & processing technologies emerged to handle this scale across many machines.

Stage 5 — Data Science Era (2010s–present)

- Statistics + machine learning + domain expertise combined on top of big-data tech
- Moves beyond describing the past (BI) toward predicting the future and prescribing optimal actions
- 'Data Scientist' emerged as a distinct professional title requiring statistics + programming + business understanding combined

Stage 6 — Agentic AI & Physical AI Era (2020s onward — emerging)

- **Agentic AI:** AI systems (often LLM-based) that autonomously plan, decide, and execute multi-step tasks with limited human supervision — given a goal, the agent queries data, analyses, interprets, and acts, chaining reasoning + actions rather than returning only a static report
- **Physical AI:** AI/ML embodied in and interacting with the physical world (robots, autonomous vehicles, smart industrial equipment) — processes real-time sensor data (vision, touch, motion) and acts back on the physical environment

Both extend the same evolutionary trend: descriptive (BI) → predictive/prescriptive (data science) → autonomous execution (agentic/physical AI). These are still emerging — treat as the current frontier, not a separate field.

Exam Tip: Write all 6 stages in order for 'evolution' questions: (1) Data Processing → (2) Data Warehousing → (3) BI → (4) Big Data (3 V's) → (5) Data Science → (6) Agentic/Physical AI. Purpose shifts: record-keeping → reporting → descriptive insight → handling scale → prediction & decision automation → autonomous, agentic & physically-embodied action.

2. Need for Data Science

Follows directly from evolution: once organisations had large, varied, fast-moving data, traditional reporting tools were no longer sufficient.

2.1 Data-side Drivers

- **Exponential growth of data:** every transaction, click, post, GPS ping, sensor reading adds up — cannot be reviewed manually, needs automated/algorithmic analysis
- **Growing variety of data:** text (reviews, tickets), images (scans, satellite), audio/video, network/graph data — needs techniques beyond simple SQL
- **Real-time / high-velocity needs:** fraud detection, ad bidding, recommendations must respond within seconds/milliseconds

2.2 Business / Organisational Drivers

- **Data-driven decision making:** pricing, hiring, stocking, lending grounded in evidence, not intuition

- **Competitive advantage:** faster/more accurate insight → act sooner than rivals
- **Personalisation at scale:** tailor products/prices/recommendations to millions of customers
- **Prediction & forecasting:** anticipate churn, equipment failure, disease outbreaks, demand spikes
- **Efficiency & automation:** automate decisions e.g. loan approval flags, delivery routing

Worked Example (Telecom Churn)

Billing DB shows some customers cancel monthly. A BI dashboard shows how many churned last month (descriptive). Data science combines call records, complaints, and payment history to predict which active customers are at high risk of churning next month, so the company can proactively offer a retention discount — insight turned into pre-emptive action.

3. Components of Data Science

Classic exam diagram: three overlapping circles (Venn diagram) — Mathematics/Statistics, Computer Science/Programming, and Domain/Business Knowledge — with Data Science at the intersection of all three. Draw this in your answer sheet.

3.1 Mathematics and Statistics

- Descriptive statistics — mean, median, mode, variance, standard deviation
- Probability theory — reasoning about uncertainty/randomness
- Inferential statistics — conclusions about a population from a sample; hypothesis testing, confidence intervals
- Linear algebra & calculus — underlies ML algorithms (vectors/matrices, optimisation)

Without this, one cannot judge if a pattern is statistically meaningful or build models that generalise to new data.

3.2 Computer Science / Programming

- Programming (Python, R, SQL) to collect, clean, transform, analyse data programmatically
- Data structures & algorithms — efficient organisation/processing of data
- Databases & big-data systems — storing/retrieving very large datasets
- Software engineering practices — reliable, maintainable, production-ready code
- Machine learning implementation — algorithms that learn patterns and predict

3.3 Domain / Business Knowledge

- Knowing which questions are worth asking and which data is relevant
- Interpreting results meaningfully for the specific field (e.g. 'high-risk' patient, 'high-value' customer)
- Recognising when a statistically valid result is impractical or misleading in practice

Domain knowledge separates data science from pure statistics or pure programming — without it, technically correct analysis can still cause poor/harmful real-world decisions.

Exam Tip: Venn diagram overlaps: Statistics + CS (no domain) = 'Machine Learning'. Statistics + Domain (no programming) = 'Traditional Research'. Programming + Domain (no statistics) = 'Danger Zone' (working software, statistically unsound conclusions). Data Science = all three together.

3.4 Supporting Sub-Components

- **Data engineering:** building pipelines to collect, move, store data
- **Data visualisation:** charts, graphs, dashboards to represent findings
- **Machine learning:** models that learn from historical data to predict/classify
- **Communication skills:** translating technical results for non-technical decision-makers

4. The Data Science Process

Also called the data science life cycle / analytics pipeline — the structured, repeatable sequence turning a business question and raw data into validated, actionable insight. Draw as a horizontal flow diagram of boxes with a feedback loop arrow (iterative, not one-directional).

4.1 Stages of the Process

1. **Problem / Question definition:** clearly define the business/research question (e.g. 'Which customers are likely to stop using our service, and why?'); relies on domain knowledge & stakeholder discussion
2. **Data collection:** gather relevant raw data — interviews, questionnaires, databases, web scraping, APIs
3. **Data cleaning / preparation:** check & fix errors, inconsistencies, missing values; reshape into usable format
4. **Data preprocessing:** integrate (combine sources), reduce, transform, discretise so data is ready for analysis
5. **Exploratory Data Analysis (EDA):** plots/graphs/summary stats to understand structure, distribution, relationships, anomalies before modelling
6. **Modelling / Analysis:** apply statistical/ML techniques to answer the question — predictive model, statistical test, clustering, etc.
7. **Evaluation / Validation:** assess performance & reliability — accuracy, generalisation to new (unseen) data
8. **Communication / Deployment:** present results via reports/visualisations/dashboards to stakeholders, or deploy model into a live system for ongoing decisions

Exam Tip: The process is *ITERATIVE, not linear*. EDA (Stage 5) may reveal a data-quality issue → loop back to Stage 3. Poor evaluation (Stage 7) → loop back to Stage 6 (different technique) or even Stage 2 (collect more data).

4.2 Why a Defined Process Matters

- Ensures consistency & repeatability across team members/projects
- Reduces risk of skipping essential steps (cleaning, validation) — a common cause of unreliable analysis
- Provides common language for technical & non-technical stakeholders to track progress
- Supports better planning — data collection & cleaning alone often consume 60–80% of total project time

4.5 Features, Advantages & Disadvantages of Data Science

4.5.1 Key Features

- Interdisciplinary by nature (statistics + programming + domain expertise)
- Works with all data types — structured, semi-structured, unstructured
- Follows a defined, iterative process (problem definition → communication/deployment)
- Goes beyond description — supports predictive & prescriptive analytics

- Heavily reliant on machine learning & statistical modelling
- Often produces reusable data products (models/pipelines generating fresh predictions continuously)

4.5.2 Advantages

- Enables prediction & forecasting — act before an event occurs
- Uncovers hidden, non-obvious patterns in large/complex datasets
- Supports personalisation at scale
- Improves accuracy & consistency of decisions (evidence-based)
- Adaptable across virtually any domain
- Can automate repetitive analytical/decision-making tasks

4.5.3 Disadvantages / Challenges

- High resource & skill requirements — costly specialists
- Time-consuming pipeline — collection & cleaning consume 60–80% of project time
- Data quality dependency — 'garbage in, garbage out'
- Model interpretability issues — 'black box' problem
- Ethical & privacy risk — potential for misuse/bias if not governed
- Infrastructure cost — significant computing/storage investment

4.6 Features, Advantages & Disadvantages of Business Intelligence

4.6.1 Key Features

- Centred on structured, internal data (ERP, CRM, transactional DBs) consolidated in a data warehouse
- Reporting & dashboarding focus — pre-built reports, scorecards, dashboards
- OLAP-based analysis — slicing/dicing along business dimensions
- Relatively short pipeline: extract → aggregate → visualise
- User-friendly for business users — little/no programming required

4.6.2 Advantages

- Fast, clear visibility into current & past performance via dashboards/KPIs
- Easier to learn & use than full data science tooling
- Faster to implement & cheaper to maintain
- Improves everyday operational decision-making
- Well-suited to standard, repeated reporting (monthly/quarterly reports)

4.6.3 Disadvantages / Limitations

- Backward-looking only — describes the past, doesn't predict the future
- Limited to mostly structured data
- Cannot automatically uncover deep/non-obvious patterns
- Dependent on well-structured, pre-integrated data (data warehouse quality)
- Limited support for automation/prescriptive action

Exam Tip: BI's limitations (backward-looking, structured-only, no auto pattern discovery) are exactly the gaps Data Science was developed to fill — link this back to Section 1.2, Stage 3 → Stage 5.

5. Difference between Data Science and Business Intelligence

5.1 The Three Levels of Analytics

1. **Descriptive analytics** — 'What happened?' Summarising historical/current data via reports & dashboards. Core territory of BI.
2. **Predictive analytics** — 'What is likely to happen?' Statistical/ML models forecasting future outcomes. Core territory of Data Science.
3. **Prescriptive analytics** — 'What should we do about it?' Recommends specific actions, often using optimisation. Most advanced territory of Data Science.

5.2 Detailed Comparison

Aspect	Business Intelligence (BI)	Data Science
Primary focus	Descriptive — what happened / what is happening now	Descriptive, predictive & prescriptive — what happened, what will happen, what to do
Type of data	Mostly structured (internal: sales, finance, ERP)	Structured, semi-structured & unstructured (text, images, sensors, web)
Techniques used	Reporting, dashboards, OLAP, SQL queries	Statistics, ML, data mining, programming, advanced analytics
Typical output	Static/interactive reports & dashboards	Predictive models, algorithms, automated systems, data products
Time orientation	Backward-looking	Forward-looking (+ historical)
Skill set	Business analysts; SQL & dashboarding tools	Statisticians, programmers, domain experts
Process complexity	Short pipeline: extract, aggregate, visualise	Full process: collection→cleaning→preprocessing→EDA→modelling→evaluation→deployment
Cost & implementation	Faster, lower cost, minimal coding	Higher cost, longer timelines, specialised skills/infra
Level of automation	Human analyst interprets & decides manually	Can automate the decision itself (fraud flagging, recommenders)

Aspect	Business Intelligence (BI)	Data Science
Example question	'What were our sales last quarter by region?'	'Which customers will churn next quarter, why, and what offer to send?'

Exam Tip: One-line exam answer: 'BI tells you what happened; Data Science tells you what happened, why it happened, what is likely to happen next, and what to do about it.'

6. Applications of Data Science in Various Fields

For exams: name the field, give 1–2 concrete applications, and state the type of data used.

6.1 Healthcare

- Disease prediction & early diagnosis (e.g. diabetes risk from patient records)
- Medical image analysis (X-rays, MRI) to assist radiologists
- Personalised treatment plans; predicting hospital readmission risk
- Optimising staff scheduling & bed allocation

6.2 Finance and Banking

- Real-time fraud detection on card transactions
- Credit risk scoring for loan approval
- Algorithmic trading strategies
- Customer segmentation; anti-money-laundering pattern detection

6.3 E-commerce and Retail

- Recommendation systems; dynamic/personalised pricing
- Demand forecasting for inventory
- Customer sentiment analysis from reviews
- Market-basket analysis (frequently bought together)

6.4 Transportation and Logistics

- Route optimisation for delivery fleets
- Predictive maintenance of vehicles
- Ride-hailing demand & surge-pricing prediction
- Perception/decision systems in autonomous/self-driving vehicles

6.5 Manufacturing

- Predictive maintenance of factory machinery (reduce downtime)
- Automated visual quality control (defect detection)
- Supply-chain optimisation (cost, speed, reliability)

6.6 Government and Public Policy

- Census/demographic analysis for policy planning
- Crime pattern prediction for policing resources
- Urban & traffic planning; disaster resource distribution
- Public-health surveillance (disease spread tracking)

6.7 Education

- Learning analytics tracking student engagement
- Personalised/adaptive learning paths
- Predicting student performance & at-risk (dropout) students

6.8 Social Media and Entertainment

- Content recommendation (video/song/post)
- Sentiment & trend analysis
- Targeted advertising based on behaviour
- Audience engagement analysis for content strategy

Exam Tip: Across all fields the same data science process (Section 4) applies — only the data, domain knowledge, and specific question change.

7. Ethical Issues in Data Science

As data science influences decisions affecting people's lives (loans, diagnoses, hiring, policing), it raises ethical questions at every stage of the process — not only at the end.

7.1 Key Ethical Concerns

- **Bias and fairness:** data/models can reflect real-world social biases (e.g. under-represented hiring data disadvantages a group again); bias can enter at collection, labelling, or modelling stage
- **Informed consent:** individuals should know when data is collected, for what purpose, and have a genuine opportunity to agree/refuse
- **Transparency and explainability:** black-box models make it hard to understand/challenge a decision
- **Accountability:** when a data-driven decision causes harm, responsibility (organisation/scientist/system) must be clear, with a route to correction
- **Data ownership:** who owns the data — the individual it describes, or the collecting organisation?
- **Misuse of insights:** patterns about habits/vulnerabilities/emotional state could be used to manipulate or exploit individuals

Illustrative Scenario

A ride-hailing app's demand-prediction model, trained only on historical data from a disadvantaged neighbourhood with low smartphone ownership, may systematically under-allocate drivers there in future — reinforcing inequality rather than correcting it. A technically 'correct' model can still be ethically questionable if data/context are not carefully considered.

Exam Tip: Ethical issues are not just a legal/compliance formality — they determine public trust in data science outputs; unaddressed failures cause real, lasting harm.

8. Data Privacy and Security

Closely related but conceptually distinct — a frequent exam question is to distinguish them clearly; do not treat as synonyms.

8.1 Data Privacy

Concerns appropriate collection, use, sharing, and disclosure of personal/sensitive information — giving individuals meaningful control over their own data.

- **Data minimisation:** collect only what is genuinely necessary for a stated purpose
- **Anonymisation / pseudonymisation:** mask identifying details so individuals can't be re-identified
- **Transparency:** clearly inform individuals what data is collected & how used
- **Individual rights:** right to access, correct, and (often) delete one's data

8.2 Data Security

Concerns protecting data from unauthorised access, alteration, loss, or destruction — from external attackers or internal misuse. Security underpins privacy: data cannot stay private if not kept secure.

- **Access control:** restrict access based on role & need ('least privilege')
- **Encryption:** protect data at rest and in transit
- **Authentication & authorisation:** verify identity, confirm permissions
- **Monitoring & auditing:** track access, detect unusual activity
- **Backup & disaster recovery:** prevent permanent data loss (hardware failure, disaster, ransomware)

8.3 Comparison

Aspect	Data Privacy	Data Security
Primary concern	Appropriate collection, use & sharing of data	Protection of data from unauthorised access or loss
Focus	Rights of the individual the data describes	Technical safeguards protecting the data itself
Typical measures	Consent, transparency, anonymisation, minimisation	Encryption, access control, authentication, monitoring
Governing question	'Should this data be used this way, and does the person know/agree?'	'Can an unauthorised party get to this data?'

Exam Tip: A dataset can be secure but not private (well-protected from hackers, yet used without consent) — and an org can have great privacy policy on paper undermined by weak security. Discuss as complementary, not interchangeable.

9. Responsible Data Use

The practical bridge between ethical principles (Section 7) and privacy/security safeguards (Section 8) — what a data scientist actually does at every stage of the process (Section 4).

9.1 Principles of Responsible Data Use

- **Purpose limitation:** use data only for the purpose originally collected, unless further explicit consent is obtained
- **Fairness:** regularly check data/models for bias and correct it
- **Transparency:** be open with stakeholders about how data is collected, processed, used
- **Accountability:** take clear ownership of outcomes, including unintended/harmful ones
- **Minimisation and proportionality:** collect/retain only what's necessary, only as long as needed
- **Respect for data subjects:** treat people in a dataset as individuals with rights, not anonymous rows

Exam Tip: *Responsible data use is NOT a one-time checklist — it's an ongoing practice embedded in every stage: collection, cleaning, preprocessing, analysis, reporting, deployment.*

Review Questions — CO-1

1. Trace the evolution of data science from the data-processing era to the present, explaining what changed at each stage, including the emerging Agentic AI and Physical AI stage.
2. What is meant by 'Agentic AI' and 'Physical AI'? How do they extend the data science process beyond prediction?
3. Explain, with examples, why data science has become necessary for modern organisations.
4. Describe the three core components of data science using the Venn-diagram model, and explain what each contributes.
5. List and explain, in order, the stages of the data science process. Why is the process considered iterative rather than linear?
6. List the key features, advantages, and disadvantages of Data Science.
7. List the key features, advantages, and disadvantages of Business Intelligence.
8. Differentiate between Business Intelligence and Data Science, using the three-levels-of-analytics framework (descriptive, predictive, prescriptive).
9. Describe three applications of data science in fields of your choice, stating the type of data used in each case.
10. What are the main ethical issues that arise in data science practice? Illustrate one with an example.
11. Distinguish between data privacy and data security, giving two practices associated with each.
12. Explain the principle of responsible data use and how it connects the ethical and privacy/security concepts studied in this unit.

Glossary of Key Terms — CO-1

Term	Definition
Data Science	Interdisciplinary field using scientific methods, statistics, algorithms and systems to extract insight from data
Data Warehouse	A central repository consolidating data from multiple systems, organised for querying and reporting
OLAP	On-Line Analytical Processing — technology for multi-dimensional analysis (slicing/dicing dashboards)
Big Data	Data characterised by high volume, velocity and variety, exceeding traditional processing capability
Descriptive Analytics	Analytics answering 'what happened' — core of BI

Term	Definition
Predictive Analytics	Analytics answering 'what is likely to happen' — core capability of data science
Prescriptive Analytics	Analytics answering 'what should we do' — most advanced level of data science
Data Privacy	Appropriate collection, use, sharing and disclosure of personal data, respecting individual rights
Data Security	Protection of data from unauthorised access, alteration, loss or destruction
Bias (data/models)	Systematic unfairness in data or model outcomes, often reflecting real-world inequalities in source data
Agentic AI	AI systems that autonomously plan, decide and execute multi-step tasks with limited human supervision
Physical AI	AI embodied in and interacting with the physical world, e.g. robots, autonomous vehicles, smart industrial equipment

Quick Revision Summary — CO-1

- **Evolution:** data processing → data warehousing → BI → big data (3 V's) → data science → agentic/physical AI (emerging)
- **Need:** exponential data growth, variety, real-time demands, data-driven decisions, competitive advantage, prediction, automation
- **Components:** statistics/mathematics + computer science/programming + domain knowledge (three-circle Venn diagram)
- **Process:** problem definition, collection, cleaning, preprocessing, EDA, modelling, evaluation, communication (iterative)
- **Data Science features/adv/disadv:** interdisciplinary, all data types, predictive power vs high cost, skill needs, black-box risk
- **BI features/adv/disadv:** structured data, dashboards, fast & easy vs backward-looking only, limited automation
- **DS vs BI:** three levels of analytics — descriptive (BI) → predictive → prescriptive (DS); also differ in cost, automation, data variety
- **Applications:** healthcare, finance, e-commerce, transportation, manufacturing, government, education, social media
- **Ethics:** bias/fairness, consent, transparency, accountability, ownership, misuse
- **Privacy vs Security:** privacy = appropriate use/sharing (consent, minimisation); security = protection from unauthorised access (encryption, access control)
- **Responsible use:** purpose limitation, fairness, transparency, accountability, minimisation, respect for data subjects